

WEEK 2 FUNCTION OPTIMIZATION

```
import random

# Function to optimize
def fitness(x):
    return x ** 2

# PSO parameters
NUM_PARTICLES = 10
ITERATIONS = 50
W = 0.5          # Inertia weight
C1 = 1.5          # Cognitive coefficient
C2 = 1.5          # Social coefficient

MIN_X = -10
MAX_X = 10

# Initialize particles
particles = []

for i in range(NUM_PARTICLES):

    position = random.uniform(MIN_X, MAX_X)
    velocity = random.uniform(-1, 1)

    particles.append({
        "position": position,
        "velocity": velocity,
        "best_position": position,
        "best_fitness": fitness(position)
    })

# Find initial global best
global_best = max(
    particles,
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    global_best = max(
        particles,
        key=lambda p: p["best_fitness"]
    )

    global_best_position = global_best["best_position"]
    global_best_fitness = global_best["best_fitness"]

# PSO iterations
for iteration in range(ITERATIONS):

    for particle in particles:

        # Update velocity
        r1 = random.random()
        r2 = random.random()

        particle["velocity"] = (
            W * particle["velocity"]
            + C1 * r1 *
                (particle["best_position"]
                 - particle["position"])
            + C2 * r2 *
                (global_best_position
                 - particle["position"])
        )

        # Update position
        particle["position"] += particle["velocity"]

        # Keep position within range
        particle["position"] = max(
            MIN_X,
            min(MAX_X, particle["position"])
        )

        # Calculate fitness
        fitness = fitness_function(particle["position"])

```

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# Calculate fitness
current_fitness = fitness(
    particle["position"]
)

# Update personal best
if current_fitness > particle["best_fitness"]:

    particle["best_position"] = (
        particle["position"]
    )

    particle["best_fitness"] = (
        current_fitness
    )

# Update global best
if current_fitness > global_best_fitness:

    global_best_position = (
        particle["position"]
    )

    global_best_fitness = (
        current_fitness
    )

# Output result
print("Best Position:", global_best_position)
print("Best Fitness:", global_best_fitness)
```

... Best Position: -10
Best Fitness: 100